



## Original Research Article

## Impact of smoking on lung cancer patterns: A public health perspective from a tertiary centre in Western India

Chirag Hasmukhbhai Chakravarti<sup>1\*</sup>, Bhavesh M Patel<sup>1</sup>, Dacharla Sai Jaswanth<sup>1</sup>, Arti Dhawal Shah<sup>1</sup>, Mothiganesh G<sup>1</sup>, Medha Jain<sup>1</sup><sup>1</sup>Dept. of Respiratory Medicine, Smt. B.K. Shah Medical Institute & Research Centre, Sumandeep Vidyapeeth Deemed to be University, Piparia, Vadodara, Gujarat, India

## Abstract

**Background:** Lung cancer is the leading cause of cancer-related mortality worldwide and continues to rise in India due to high tobacco use, occupational exposures, and environmental pollution. While adenocarcinoma has become the predominant subtype in many regions globally, Indian studies historically show squamous cell carcinoma (SCC) dominance due to widespread bidi smoking. With changing lifestyle and smoking patterns, updated regional data are essential.

**Aim:** To evaluate the impact of smoking on demographic, clinical, and histopathological patterns of lung cancer in a tertiary care centre in Western India.

**Materials and Methods:** A cross-sectional observational study was conducted from January 2022 to January 2024, including 100 consecutive, consenting adults with radiologically and histopathological confirmed lung cancer. Clinical features, smoking history, biopsy modality, and tumour histology were analysed. Statistical associations between smoking status and histological subtype were assessed using chi-square tests, with odds ratios (OR) and 95% confidence intervals (CI).

**Results:** Among 100 patients, 76% were smokers and 24% non-smokers. Males accounted for 85%. The mean age was  $59.8 \pm 9.7$  years (range: 34–79). Common symptoms included cough (89%), breathlessness (84%), chest pain (75%), and weight loss (71%). The most frequent histological subtype was adenocarcinoma (39%), followed by SCC (38%), poorly differentiated carcinoma (10%), SCLC (6%), and LCC (2%). Among smokers, SCC predominated (35/76; 46.1%), whereas adenocarcinoma was most common among non-smokers (12/24; 50%). Smoking was significantly associated with SCC ( $\chi^2 = 6.42$ ;  $p = 0.011$ ; OR = 2.68, 95% CI 1.22–5.87). Bronchoscopy-guided biopsy was the most frequently performed diagnostic modality (61%).

**Conclusion:** Adenocarcinoma is emerging as the predominant lung cancer subtype overall, yet SCC remains strongly associated with tobacco smoking. High rates of advanced-stage presentation highlight the need for improved early detection, reduction in misdiagnosis as tuberculosis, and strengthened tobacco control measures. Public health interventions aimed at smoking cessation and awareness are urgently required to curb the rising burden of lung cancer.

**Keywords:** Lung cancer, Smoking, Adenocarcinoma, Squamous cell carcinoma, Histopathology, Tobacco.

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## 1. Introduction

Lung cancer remains a major global health challenge due to its high incidence, late diagnosis, and poor overall survival. According to GLOBOCAN 2018, lung cancer contributed to 2.1 million new cases and 1.8 million deaths worldwide, ranking first among cancer-related deaths.<sup>1-5</sup> In India, lung cancer accounts for 5.9% of all malignancies, with a disproportionately higher burden among males.<sup>6,7</sup>

## 1.1. Changing histopathological landscape

Historically, squamous cell carcinoma (SCC) was the most common lung cancer subtype in India, attributed to the prevalence of bidi smoking, which delivers high concentrations of tar and carcinogens.<sup>11</sup> However, recent Indian studies indicate a shift toward adenocarcinoma, paralleling global trends.<sup>4,6</sup> This transition may relate to:

1. Increased use of filtered cigarettes,

\*Corresponding author: Chirag Hasmukhbhai Chakravarti  
Email: [jaswanth.3780@gmail.com](mailto:jaswanth.3780@gmail.com)

2. Deeper inhalation patterns,
3. Rising air pollution (PM2.5 and biomass fuel),
4. Improved radiological detection of peripheral lesions, and
5. Changing biological behaviour of tumours.

### 1.2. The smoking–cancer connection

Tobacco use contributes to nearly 85% of bronchogenic carcinoma cases.<sup>8-13</sup> Smoke-derived carcinogens such as polycyclic aromatic hydrocarbons promote DNA damage, driving SCC especially in central airways. Conversely, adenocarcinoma is more common among non-smokers and women, and has notable genetic driver mutations (EGFR, ALK, ROS1).

### 1.3. Diagnostic challenges in India

A significant barrier in India is the frequent misdiagnosis of lung cancer as tuberculosis, especially when patients present with chronic cough, weight loss, and radiographic consolidation [8]. This diagnostic delay results in a higher proportion of advanced-stage cancers at presentation.

### 1.4. Rationale

Given evolving smoking trends, environmental exposures, and diagnostic capabilities, updated regional data are required to:

1. Assess changing histological patterns,
2. Evaluate smoking as a risk factor,
3. Prevent diagnostic delays, and
4. Guide public health interventions.

This study therefore examines the impact of smoking on lung cancer patterns in a tertiary care centre in Western India.

## 2. Materials and Methods

### 2.1. Study design and setting

A hospital-based, cross-sectional observational study was conducted at the Department of Respiratory Medicine, SBKS MI & RC, Vadodara.

### 2.2. Study period

January 2022 – January 2024 (24 months).

### 2.3. Study population

One hundred consecutive adult patients (>18 years) with radiologically and histopathological confirmed primary lung cancer.

### 2.4. Inclusion criteria

1. Adults presenting with clinical suspicion of lung cancer.
2. Radiological features suggestive of malignancy.
3. Histopathological proven lung cancer.
4. Provided informed written consent.

### 2.5. Exclusion criteria

1. Previously diagnosed lung cancer cases.
2. HIV or HBsAg positivity.
3. Patients refusing consent.

### 2.6. Data collection

#### 2.6.1. Variables included

1. Age, sex, smoking status, smoking type (bidi/cigarette),
2. Symptom profile,
3. Imaging findings,
4. Biopsy modality (bronchoscopy, CT-guided, USG-guided),
5. Histopathological diagnosis.

### 2.7. Statistical analysis

1. Data analysed using SPSS v26.
2. Descriptive statistics: mean, SD, frequencies, percentages.
3. Associations between smoking and histological subtype evaluated using chi-square test or Fisher's exact test.
4. Odds ratios (OR) and 95% confidence intervals (CI) calculated.
5.  $p < 0.05$  considered statistically significant.

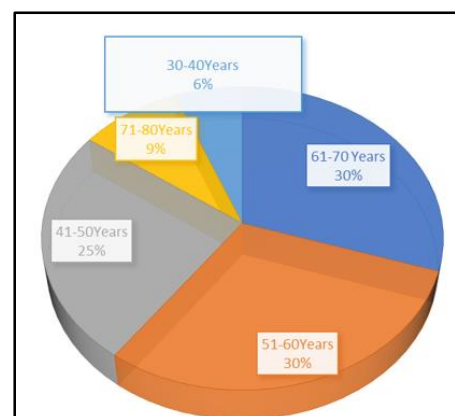
### 2.8. Ethical considerations

1. Approved by the Institutional Ethics Committee (IEC number will need to be inserted by you).
2. Written informed consent obtained from all participants.
3. No conflicts of interest.
4. No external funding received.

## 3. Results

### 3.1. Demographic characteristics

Most common age groups (**Figure 1**):



**Figure 1:** Age distribution

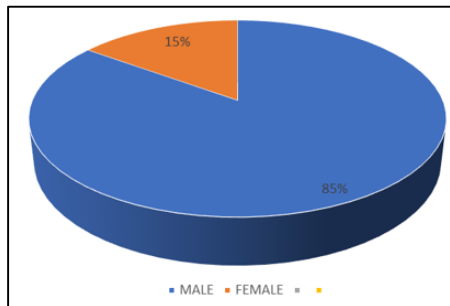
1. 61–70 years: 30%
2. 51–60 years: 30%
3. 41–50 years: 25%

4. 71–80 years: 9%
5. 30–40 years: 6%

In our study Mean age group was  $59.8 \pm 9.7$  years (range 34–79). Smokers were significantly older than non-smokers (mean age: 61.3 vs 55.1 years)

### 3.2. Gender distribution (Figure 2)

1. Male: 85 (85%)
2. Female: 15 (15%)



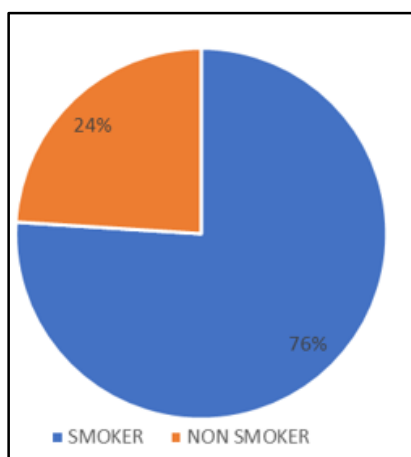
**Figure 2:** Gender distribution

### 3.3. Smoking was predominantly a male habit in this study

1. Male smokers: 71/85 (83.5%)
2. Female smokers: 5/15 (33.3%)

### 3.4. Among 100 patients (Figure 3)

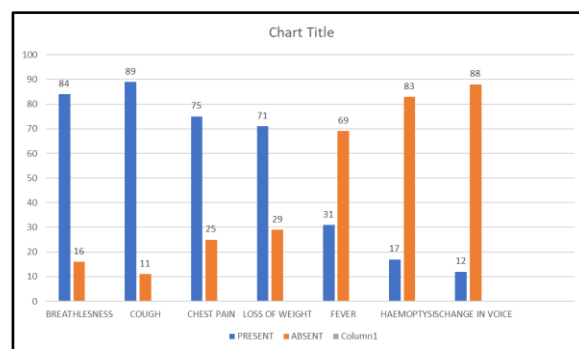
1. Smokers: 76 (76%)
2. Non-smokers: 24 (24%)
3. Most smokers used bidi, followed by cigarettes.



**Figure 3:** Habits

### 3.5. Common Symptoms

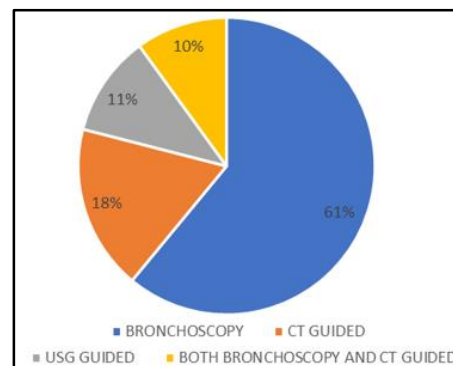
Smokers had higher frequencies of cough, expectoration, and hemoptysis, while non-smokers more often presented with constitutional symptoms such as weight loss and fever. (Figure 4)



**Figure 4:** Common symptoms

### 3.6. Type of biopsy

Bronchoscopy was more commonly utilized for central lesions (mostly SCC), whereas CT-guided biopsy was preferred for peripheral lesions, typical of adenocarcinoma. (Figure 5)

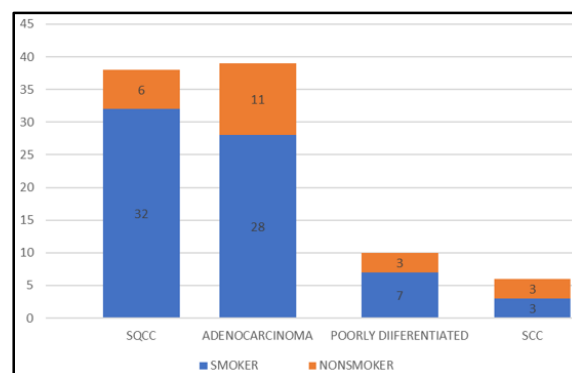


**Figure 5:** Type of biopsy

### 3.7. Type of lung carcinoma

Adenocarcinoma was the most common subtype overall, slightly exceeding SCC. (Figure 6)

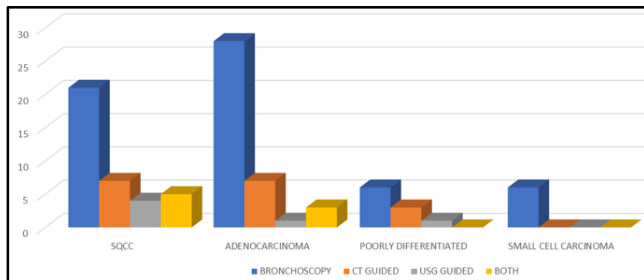
1. SCC was nearly twice as common in smokers compared to non-smokers (46.1% vs 25%).
2. Adenocarcinoma was more common in non-smokers (50% vs 38.1%).
3. SCLC showed no strong relationship with smoking in this sample.



**Figure 6:** Most common type of carcinoma

### 3.9. Statistical association between smoking and histology

Smokers had 2.68 times higher odds of developing SCC compared to non-smokers, showing a statistically significant association. (Figure 7)



**Figure 7:** Common types of Biopsies in various histopathological types

### 3.10. Summary of results

1. Tumour location
  - a. SCC more frequently central → bronchoscopy diagnostic yield high.
  - b. Adenocarcinoma peripheral → CT/USG-guided biopsy more useful.
2. Gender-based histology
  - a. Women (non-smokers) → predominantly adenocarcinoma.
  - b. Men (mostly smokers) → predominantly SCC.
3. Symptom differences
  - a. Smokers: cough, expectoration, hemoptysis
  - b. Non-smokers: fatigue, fever, weight loss

## 4. Discussion

Globally, lung cancer remains the leading cause of cancer mortality, as highlighted by the GLOBOCAN 2018 estimates reporting 1.8 million deaths annually.<sup>1</sup> Our findings reaffirm this burden and emphasize the growing relevance of lung cancer in India, where incidence continues to rise, consistent with the NCRP-ICMR report.<sup>2</sup>

Historically, India reported a predominance of squamous cell carcinoma (SCC), largely due to widespread bidi smoking.<sup>3</sup> However, several studies have recently documented a shift toward adenocarcinoma (ADC). Malik et al. reported increasing ADC cases at AIIMS, suggesting a changing landscape in tumour biology in India.<sup>4</sup> Noronha et al. also observed the rising incidence of ADC, particularly among women and non-smokers.<sup>10</sup> Our study concurs with these observations: although SCC remains dominant among smokers, adenocarcinoma (39%) has become the overall most common subtype, suggesting that India is transitioning toward patterns previously seen in Western populations.

The shift may reflect changes in the type of smoking products (filter cigarettes vs bidis), increased exposure to indoor/outdoor pollution, and improved diagnostic imaging capable of detecting peripheral lesions typical of adenocarcinoma. Consistent with the well-established

carcinogenicity of tobacco smoke described by Hecht et al.<sup>5</sup> and WHO/IARC evaluations,<sup>7-10</sup> our study demonstrates a robust association between smoking and SCC, with smokers exhibiting 2.68-fold higher odds of developing SCC ( $p = 0.011$ ). This aligns with classical Indian studies by Jindal and by Behera & Balamugesh, both of whom emphasized the strong causal relationship between tobacco and SCC.<sup>3,9</sup>

Our data also compare well with findings from North India where chronic inhalation of bidi smoke was shown to predispose to central airway tumours, particularly SCC]. International cohorts, including, also document a strong dose-dependent association between active smoking and SCC, strengthening the biological plausibility of our observations.<sup>11-14</sup>

Interestingly, adenocarcinoma occurred more commonly in non-smokers (50%), consistent with global literature showing that ADC is increasingly diagnosed in women, non-smokers, and individuals exposed to air pollution or carrying driver mutations (EGFR, ALK). Such patterns were also observed by Singh et al.<sup>6</sup> and Sharma et al.<sup>18</sup>

The symptom profile in our study—dominated by cough (89%), breathlessness (84%), and chest pain (75%)—is comparable with multiple Indian studies including.<sup>1</sup> However, an important clinical observation in our study mirrors those of substantial proportion of patients presented with symptoms that overlap heavily with tuberculosis (TB). Misdiagnosis of lung cancer as TB remains a critical problem in India due to high TB prevalence. This leads to treatment delays, disease progression, and poorer survival outcomes.<sup>8</sup>

Thus, our study reinforces the need for high clinical suspicion, especially when older smokers present with persistent respiratory symptoms unresponsive to empirical anti-TB therapy.

Our study demonstrates the dominance of bronchoscopy-guided biopsy (61%), reflecting the high prevalence of centrally located tumours such as SCC among smokers. This is consistent who noted a higher diagnostic yield for bronchoscopy in SCC, whereas CT-guided biopsy is often preferred for peripheral lesions such as ADC. The correct selection of biopsy method is essential for timely diagnosis, tissue adequacy, and molecular testing. With the increasing global emphasis on immunotherapy and targeted therapy, accurate and early tissue diagnosis is becoming more crucial than ever.<sup>2</sup>

Taken together, our results confirm multiple emerging epidemiological shifts in India:

1. Transition from SCC to ADC as the dominant subtype, like global patterns.
2. SCC remains tightly linked to smoking, reinforcing the public-health need for tobacco regulation.

3. Adenocarcinoma is increasing among non-smokers, especially women—a pattern supported by Noronha et al.<sup>8</sup>
4. Air pollution and genetic driver mutations may contribute to rising ADC rates, as suggested in international studies.<sup>19</sup>
5. Diagnostic delays and TB misclassification remain persistent challenges in India.<sup>10</sup>
6. These insights are essential for designing screening strategies, public health interventions, and diagnostic algorithms specific to Indian populations.

This study is significant and merits publication due to the following strengths:

1. Provides Updated Western India Data (2022–2024) Very few recent Indian studies cover the post-COVID era where smoking patterns, pollution exposure, and healthcare-seeking behaviours have shifted.
2. Strong Comparative Analysis of Smokers vs Non-Smokers
3. The study clearly demonstrates histological variations, with statistical confirmation ( $p = 0.011$ ). This is critical evidence for targeted public-health messaging.
4. Highlights the Emerging Dominance of Adenocarcinoma. This shift has important implications for molecular diagnostics, as ADC often requires EGFR/ALK testing for targeted therapies.
5. Uses Multiple Biopsy Modalities Provides practical insights into biopsy technique selection based on tumour location.
6. Addresses the Ongoing Issue of Misdiagnosis with Tuberculosis. A uniquely Indian problem with significant morbidity.
7. Contributes to Tobacco Control and Policy The study provides real-time evidence supporting national anti-tobacco initiatives.

#### 5.1. A. Public health impact

The data directly support India's National Tobacco Control Programme (NTCP) and can guide region-specific interventions to curb smoking and prevent SCC.

#### 5.2. B. Clinical relevance

Understanding histological shifts helps clinicians select appropriate biopsy methods, consider early molecular testing, and avoid misdiagnosis.

#### 5.3. C. Academic contribution

This manuscript fills a critical gap in Western Indian epidemiology, providing post-pandemic lung cancer patterns that have not previously been documented.

#### 5.4. D. Research value

The comparative smoker vs non-smoker analysis, backed by statistical significance, offers strong evidence suitable for future meta-analyses and policy recommendations.

### 5. Limitations

1. Single-centre design.
2. Molecular analysis not included.
3. Stage-wise distribution not assessed.

#### 6.1. Public health implications

1. Stronger tobacco control is urgently needed, especially targeting bidi users.
2. Clinician training must emphasise differentiating lung cancer from TB.
3. Awareness campaigns should address non-smoker lung cancer risks (air pollution, biomass smoke).

### 6. Conclusion

In summary, this study contributes meaningful, contemporary, and regionally relevant insights into the evolving patterns of lung cancer in India. By demonstrating the strong association of smoking with SCC and the rising prevalence of adenocarcinoma among non-smokers, it underscores the dual necessity of tobacco control and improved diagnostic vigilance. These findings make the study a valuable addition to Indian literature and a strong candidate for publication in medical journals focusing on pulmonology, oncology, and public health.

### 7. Source of Funding

None.

### 8. Conflict of Interest

None.

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